

oueees-202606 topic 00

電気工学特別講義

2026年6月23日分 イントロダクション

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23-JUN-2026

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On the internet

@jj1bdx

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Lecture notes and reporting

- <https://github.com/jj1bdx/oueees-202606-public/>
- Check out the README.md file and the issues!
- Keyword at the end of the talk
- URL for submitting the report at the end of the talk

2026: a year of uncertainty and hype

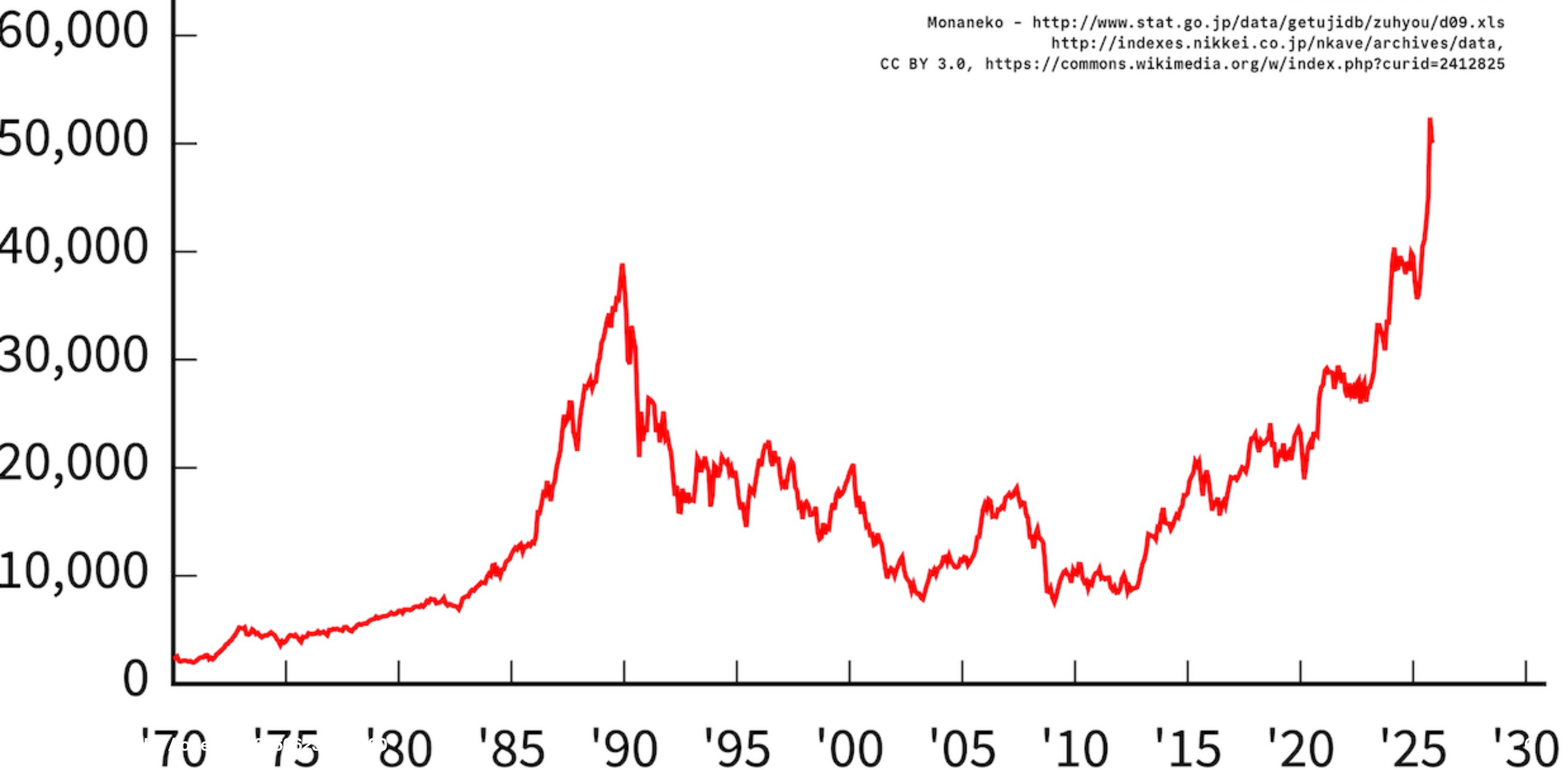
- Oil supply shortage since 1973, *53 years ago*
- 12 years since Russo-Ukrainian war began in 2014
- 6 years since COVID-19 started in 2020
- Soaring Nikkei 225 index (~35 years since the bubble economy in 1980s)
- IPOs of major AI companies: OpenAI (ChatGPT) and Anthropic (Claude)

Index (JPY)

Nikkei 225 Index 1970-2026

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Monaneko - <http://www.stat.go.jp/data/getujidb/zuhyou/d09.xls>
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2026: the hard-cold reality of color ink shortage ¹



¹ カルビー株式会社, "中東情勢の影響による一部商品仕様見直しのお知らせ", May 12, 2026,

War zone everywhere

- Hormuz Strait and USA-Iran war since March 2026
- Gaza Strip has become a wasteland since 2023
- Conflict escalation everywhere in the world
- *Japan is not an exception*

Japan has already engaged in the wars

- Japan has most deeply committed in the military and peace-keeping activities on regional conflicts since 1952
- Always remember: *open and fair trade is the only way to survive for a country like Japan which does not have sufficient natural resources*

How can we survive in Japan?

- **The levels of tension in the regional conflicts which Japan is engaged in are rising to much higher levels than those in the cold war during 1945-1991**
- **Japan started the Pacific War because of oil embargo in 1941**

We need to change our perspectives

- We are under the process of revising our understating of how the world is built and formed**
- It's not just about the economics and technology**
- It's more about the scarcity of resources and redistribution**

Infrastructure crises

- Broken supply chains: crude oil, rare-earth minerals, natural gas
- Fragile supply chains: food and semiconductors
- Global warming and energy and water shortage

Are we heading into the apocalypse?

Logistics is digital

- **Physical ubiquity and software world are co-dependent**
- **We need ample supplies of daily necessities to keep AI running**
- **Can we maintain this?**

We need to survive, anyway

- Staying alive is hard**
- Safety first is precious**
- Getting out of slavery is extremely difficult**

We need to prevent slavery

Slavery is everywhere



Slavery is everywhere

- Cruelty and inequality going on everywhere by people with power to enslave oppressed and marginalized people
- *Who are the marginalized people? You can be one at any time*

So what can we do?

- Technology is for the people, *not for the billionaires, and never for the oppressive organizations*
- Can we keep our technology free?

Who I am / 自己紹介

- Software engineer, *formerly called himself Professional Internet Engineer*
- 技術士（情報工学部門）
- 力武健次技術士事務所 所長
- 情報処理安全確保支援士

My research work

- Guest Researcher, Pepabo R&D Institute, GMO Pepabo, Inc.
- GMOペパボ株式会社 ペパボ研究所 客員研究員



Pepabo R&D Institute, GMO Pepabo, Inc.

ペパボ研究所

by GMOペパボ

My career

- Erlang/OTP, Elixir, C, VAX/VMS now OpenVMS, FreeBSD, Linux, TCP/IP, PHP, mruby, Lua, C++, C#, Visual Studio, Moodle, macOS, Windows, Vim, VS Code, Arduino, AVR, radio engineering, music, distributed systems, fault tolerance, software defined radio, Python, R, signal processing, x86_64 AVX, HTML, CSS, JavaScript, TypeScript, adaptive filter, machine learning / deep learning, LLM and AI apprentice, Claude Code prompting, Kermit protocol, whatever
- 36 years in Computer Science, 21 years since PhD, 50 years of ham radio op as @jj1bdx, 2010-2012: Professor, ACCMS/IIMC, Kyoto University, ACM Senior Member, amateur DJ and musician, whatever

My DJ appearance at Room 303 Radio, WPÜ Shinjuku, Shinjuku City,
Tokyo, Japan on May 30, 2026 (quoted from my Instagram post)



<https://www.instagram.com/p/DZBgkzMNPht/>

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bdx

bdx_tokyo My first-ever public DJ play at @room303radio on May 30, 2026. Thanks, @mattlecler, for the kind assistance! The song here: "Elementary Particle" by Akiyama, from Neuro Net Recordings.

1w See translation



Liked by room303radio and 8 others
June 1



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My recent career is not a brilliant one, though

- I've got laid off on June 2013
- >13 years of ongoing survival
- as an independent IT consultant
- and a *software engineer*

Past records are meaningless

- You need to live your life and earn money
- ... then you can work on what you really want to do

Ignore past achievements
Focus on *now*

Ignore everybody
to stay creative and maintain originality

Information delivery on internet

— Our lecture theme

How internet works

(In other words)

容錯設計

Fault-tolerant design

Internet is a survival technology

Modern life is full of failures

How internet works under failures?

Technology 1:

Packet switching

Technology 2:

Flexible packet routing

Technology 3:

Centralization and decentralization

Topic sections (1/3)

- **Latency and Laws of Physics**
- **Centralized communication**
- **Multiplexing**
- **Packet switching**
- **Routing basics**

Topic sections (2/3)

- **IP addresses**
- **Routing in details**
- **Network transports**
- **Cloud computing basics**
- **Social implication of cloud computing**

Topic sections (3/3)

- **Network fault-tolerance**
- **Network services and programming trends**
- **Wireless/radio and internet**
- **Information warfare and radio surveillance**
- **"Artificial Intelligence" and the reality**
- **Reference books**
- **Career choice**

Summary:

- **Divide data into packets**
- **Route flexibly and wisely**
- **Decentralize and distribute**

Let's get down to business!

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